UTHO SmartCloud Optimizer

Resource Inventory and Data Source Documentation

Report generated: 24 Oct 2026, 21:47 IST
Dataset: 100 synthetic virtual machines, seed 20261023

Synthetic demo data

Every resource, metric and price in this report is generated for demonstration and testing.
None of it was retrieved from a live cloud provider, and none of it is billing information.

1. Where the resource data comes from

Our resource inventory is generated by a deterministic synthetic dataset generator (server/synth.py). It simulates the infrastructure of a fictional SaaS company. The dataset represents virtual machines and related storage and public IP resources across simulated regions, environments and teams.

How the dataset is generated:

- The generator is deterministic. It uses a seeded random number generator with seed 20261023, so running it again with the same seed produces identical data.
- The inventory holds 100 virtual machines: 12 named demo machines, plus 88 machines drawn from a seeded mix of 10 workload patterns.
- Attached resources: 100 attached storage volumes and 19 attached public IPs. Detached (orphan) resources: 4 volumes and 3 public IPs.
- Utilization measurements are modeled at 15-minute intervals for 30 days, from 23 Sep 2026 to 23 Oct 2026. Each resource has 2,880 expected intervals.
- Named demo machines carry scenario notes that explain the situation they model, for example underused development machines or production machines with peak-hour load.

Resource attributes (CPU and memory capacity, storage, environment, state, ownership and utilization patterns) are generated or assigned by the project's data-generation logic. Prices are illustrative inputs defined in the project's price book (server/config.py). They are not live cloud-provider quotes.

Distinction: the resource inventory is generated. The utilization measurements are generated from those workload patterns. The prices are illustrative assumptions applied to the generated resources.

2. Inventory summary

Resource count: 100 virtual machines. Each has a unique ID, a name, a size from the VM catalog, a region, an environment, a team, a criticality rating and a state.

By environment

Value	Resources
Development	41
Staging	30
Production	29

By region

Value	Resources
Mumbai	34
Hyderabad	33
Singapore	33

By team

Value	Resources
Platform	26
Analytics	20
Data	20
Engineering	19
QA	15

By criticality

Value	Resources
Medium	44
Low	27
High	23
Critical	6

By state

Value	Resources
running	100

By idle state

Value	Resources
active	66
idle_outside_hours	23
consistently_idle	5
temporarily_idle	5
insufficient_data	1

By VM size (catalog)

Size	Family	vCPU	RAM GB	Hourly INR	Resources
gp-4x16	general	4	16	7.60	28
cpu-4x8	compute	4	8	6.80	20
gp-8x32	general	8	32	15.20	17
cpu-2x4	compute	2	4	3.40	9
gp-2x8	general	2	8	3.80	7
cpu-8x16	compute	8	16	13.60	6
mem-4x32	memory	4	32	9.20	6
mem-8x64	memory	8	64	18.40	6
mem-2x16	memory	2	16	4.60	1

Attachments

Item	Count
Attached storage volumes	100
Attached public IPs	19
Detached (orphan) volumes	4
Detached (orphan) public IPs	3

3. Resource inventory table

All 100 resources in the loaded dataset. The same records are available as CSV at /api/export/resources.csv. Monthly INR is the estimated compute, storage and public IP cost for the resource, using the price book.

ID	Name	Type	Region	Environment	Team	vCPU	RAM	State	Hourly	Monthly
vm-0001	dev-sandbox-api-01	gp-4x16	Mumbai	Development	Engineering	4	16	running	7.60	7,176.90
vm-0002	prod-checkout-api-01	gp-8x32	Mumbai	Production	Platform	8	32	running	15.20	16,716.71
vm-0003	staging-reporting-mem-01	mem-4x32	Hyderabad	Staging	Data	4	32	running	9.20	8,391.14
vm-0004	dev-ml-train-01	cpu-4x8	Hyderabad	Development	Data	4	8	running	6.80	6,731.47
vm-0005	staging-web-01	gp-4x16	Singapore	Staging	Platform	4	16	running	7.60	11,776.05
vm-0006	prod-pipeline-ingest-01	mem-8x64	Mumbai	Production	Data	8	64	running	18.40	26,201.88
vm-0007	dev-qa-regression-01	gp-2x8	Mumbai	Development	QA	2	8	running	3.80	4,854.29
vm-0008	batch-reconcile-01	mem-4x32	Hyderabad	Production	Analytics	4	32	running	9.20	9,737.23
vm-0009	staging-bursty-api-01	cpu-8x16	Mumbai	Staging	Platform	8	16	running	13.60	13,329.09
vm-0010	qa-legacy-idle-01	gp-2x8	Singapore	Development	QA	2	8	running	3.80	3,268.41
vm-0011	dev-analytics-temp-01	gp-4x16	Hyderabad	Development	Analytics	4	16	running	7.60	7,140.98
vm-0012	prod-reports-storage-01	gp-4x16	Mumbai	Production	Data	4	16	running	7.60	8,520.59
vm-0013	stg-moderate-013	gp-4x16	Singapore	Staging	Engineering	4	16	running	7.60	11,327.50
vm-0014	prod-prod-steady-014	gp-8x32	Singapore	Production	Analytics	8	32	running	15.20	15,221.98
vm-0015	prod-prod-steady-015	gp-8x32	Hyderabad	Production	Engineering	8	32	running	15.20	16,702.41
vm-0016	dev-steady-low-016	gp-2x8	Singapore	Development	QA	2	8	running	3.80	4,432.07
vm-0017	stg-mem-bound-017	mem-4x32	Hyderabad	Staging	Analytics	4	32	running	9.20	9,008.22
vm-0018	dev-idle-018	gp-4x16	Mumbai	Development	Platform	4	16	running	7.60	6,106.47
vm-0019	dev-dev-office-019	cpu-4x8	Mumbai	Development	Platform	4	8	running	6.80	6,215.43
vm-0020	dev-dev-office-020	cpu-2x4	Hyderabad	Development	Engineering	2	4	running	3.40	3,358.03
vm-0021	stg-bursty-021	cpu-4x8	Mumbai	Staging	Data	4	8	running	6.80	7,329.25
vm-0022	prod-batch-022	mem-8x64	Hyderabad	Production	Platform	8	64	running	18.40	18,304.68
vm-0023	prod-prod-steady-023	gp-8x32	Singapore	Production	Analytics	8	32	running	15.20	16,923.47
vm-0024	stg-mem-bound-024	mem-4x32	Hyderabad	Staging	QA	4	32	running	9.20	7,400.94
vm-0025	dev-temp-idle-025	gp-4x16	Hyderabad	Development	QA	4	16	running	7.60	7,022.77
vm-0026	dev-steady-low-026	gp-8x32	Hyderabad	Development	Platform	8	32	running	15.20	12,071.17
vm-0027	dev-steady-low-027	gp-8x32	Hyderabad	Development	QA	8	32	running	15.20	12,524.55
vm-0028	stg-moderate-028	gp-4x16	Singapore	Staging	Engineering	4	16	running	7.60	10,918.82
vm-0029	stg-moderate-029	gp-4x16	Singapore	Staging	Engineering	4	16	running	7.60	10,919.45
vm-0030	prod-prod-peaks-030	gp-8x32	Singapore	Production	Platform	8	32	running	15.20	15,607.04
vm-0031	dev-dev-office-031	cpu-2x4	Mumbai	Development	Engineering	2	4	running	3.40	3,407.70
vm-0032	stg-moderate-032	cpu-4x8	Singapore	Staging	Analytics	4	8	running	6.80	11,651.27
vm-0033	prod-batch-033	mem-4x32	Singapore	Production	Platform	4	32	running	9.20	10,360.56
vm-0034	stg-bursty-034	cpu-8x16	Mumbai	Staging	Data	8	16	running	13.60	13,597.39
vm-0035	stg-mem-bound-035	mem-4x32	Mumbai	Staging	Analytics	4	32	running	9.20	7,312.93
vm-0036	prod-batch-036	mem-8x64	Singapore	Production	Engineering	8	64	running	18.40	16,578.17
vm-0037	dev-dev-office-037	cpu-2x4	Singapore	Development	Data	2	4	running	3.40	3,119.12
vm-0038	stg-moderate-038	gp-4x16	Singapore	Staging	Analytics	4	16	running	7.60	8,464.93
vm-0039	dev-steady-low-039	gp-2x8	Singapore	Development	Platform	2	8	running	3.80	4,119.03
vm-0040	stg-moderate-040	gp-4x16	Mumbai	Staging	Data	4	16	running	7.60	11,891.12
vm-0041	dev-dev-office-041	cpu-2x4	Mumbai	Development	Platform	2	4	running	3.40	3,107.72
vm-0042	prod-prod-steady-042	gp-8x32	Mumbai	Production	Data	8	32	running	15.20	16,433.28
vm-0043	stg-moderate-043	gp-4x16	Hyderabad	Staging	Engineering	4	16	running	7.60	8,032.98
vm-0044	stg-moderate-044	gp-4x16	Mumbai	Staging	Engineering	4	16	running	7.60	9,881.27
vm-0045	prod-prod-peaks-045	gp-8x32	Hyderabad	Production	Platform	8	32	running	15.20	19,614.21
vm-0046	dev-dev-office-046	cpu-4x8	Singapore	Development	Platform	4	8	running	6.80	6,231.51
vm-0047	stg-mem-bound-047	mem-2x16	Hyderabad	Staging	Data	2	16	running	4.60	4,854.27
vm-0048	dev-dev-office-048	cpu-2x4	Singapore	Development	Analytics	2	4	running	3.40	3,761.81
vm-0049	prod-prod-steady-049	gp-4x16	Mumbai	Production	Analytics	4	16	running	7.60	11,396.31
vm-0050	stg-moderate-050	cpu-4x8	Hyderabad	Staging	QA	4	8	running	6.80	9,820.86
vm-0051	prod-prod-steady-051	gp-4x16	Hyderabad	Production	Platform	4	16	running	7.60	12,425.76
vm-0052	dev-dev-office-052	cpu-2x4	Singapore	Development	Platform	2	4	running	3.40	3,123.28
vm-0053	dev-idle-053	gp-2x8	Singapore	Development	Data	2	8	running	3.80	4,312.08
vm-0054	prod-prod-steady-054	gp-8x32	Singapore	Production	Platform	8	32	running	15.20	17,854.21
vm-0055	prod-prod-peaks-055	cpu-8x16	Singapore	Production	QA	8	16	running	13.60	14,239.89
vm-0056	prod-batch-056	mem-8x64	Singapore	Production	Engineering	8	64	running	18.40	20,081.41
vm-0057	stg-bursty-057	cpu-8x16	Hyderabad	Staging	Engineering	8	16	running	13.60	13,566.09
vm-0058	dev-steady-low-058	gp-4x16	Mumbai	Development	Data	4	16	running	7.60	6,659.15

ID	Name	Type	Region	Environment	Team	vCPU	RAM	State	Hourly	Monthly
vm-0059	stg-moderate-059	cpu-4x8	Mumbai	Staging	Data	4	8	running	6.80	8,146.24
vm-0060	dev-dev-office-060	cpu-4x8	Mumbai	Development	Platform	4	8	running	6.80	6,218.10
vm-0061	stg-moderate-061	cpu-4x8	Singapore	Staging	Platform	4	8	running	6.80	7,610.13
vm-0062	prod-prod-steady-062	gp-8x32	Mumbai	Production	Analytics	8	32	running	15.20	16,492.04
vm-0063	stg-moderate-063	cpu-4x8	Mumbai	Staging	QA	4	8	running	6.80	8,116.03
vm-0064	dev-dev-office-064	cpu-2x4	Singapore	Development	Platform	2	4	running	3.40	3,373.89
vm-0065	stg-moderate-065	gp-4x16	Hyderabad	Staging	Data	4	16	running	7.60	10,459.08
vm-0066	dev-steady-low-066	gp-8x32	Singapore	Development	Platform	8	32	running	15.20	12,588.67
vm-0067	prod-prod-peaks-067	cpu-8x16	Mumbai	Production	Data	8	16	running	13.60	15,409.70
vm-0068	stg-moderate-068	cpu-4x8	Hyderabad	Staging	Analytics	4	8	running	6.80	9,164.30
vm-0069	prod-prod-steady-069	gp-4x16	Hyderabad	Production	Platform	4	16	running	7.60	11,133.99
vm-0070	stg-bursty-070	cpu-4x8	Singapore	Staging	QA	4	8	running	6.80	9,221.41
vm-0071	dev-dev-office-071	cpu-4x8	Singapore	Development	Analytics	4	8	running	6.80	6,672.95
vm-0072	dev-steady-low-072	gp-8x32	Mumbai	Development	QA	8	32	running	15.20	12,387.63
vm-0073	dev-idle-073	gp-4x16	Hyderabad	Development	Engineering	4	16	running	7.60	6,453.69
vm-0074	dev-steady-low-074	gp-4x16	Singapore	Development	Platform	4	16	running	7.60	7,294.31
vm-0075	dev-temp-idle-075	gp-8x32	Hyderabad	Development	Analytics	8	32	running	15.20	13,224.83
vm-0076	stg-bursty-076	cpu-4x8	Hyderabad	Staging	Platform	4	8	running	6.80	9,413.26
vm-0077	prod-prod-steady-077	gp-4x16	Mumbai	Production	Engineering	4	16	running	7.60	11,409.03
vm-0078	dev-dev-office-078	cpu-2x4	Mumbai	Development	QA	2	4	running	3.40	3,603.24
vm-0079	prod-batch-079	mem-8x64	Hyderabad	Production	Analytics	8	64	running	18.40	18,233.37
vm-0080	dev-dev-office-080	cpu-4x8	Mumbai	Development	Engineering	4	8	running	6.80	6,437.34
vm-0081	dev-dev-office-081	cpu-4x8	Singapore	Development	Analytics	4	8	running	6.80	6,311.70
vm-0082	dev-steady-low-082	gp-8x32	Mumbai	Development	Analytics	8	32	running	15.20	11,601.63
vm-0083	dev-dev-office-083	cpu-2x4	Mumbai	Development	Platform	2	4	running	3.40	4,505.58
vm-0084	dev-temp-idle-084	gp-4x16	Singapore	Development	Engineering	4	16	running	7.60	6,978.66
vm-0085	dev-steady-low-085	gp-4x16	Mumbai	Development	Data	4	16	running	7.60	6,531.57
vm-0086	dev-idle-086	gp-2x8	Singapore	Development	Data	2	8	running	3.80	4,158.35
vm-0087	prod-prod-peaks-087	cpu-8x16	Hyderabad	Production	Data	8	16	running	13.60	14,051.87
vm-0088	dev-dev-office-088	cpu-4x8	Hyderabad	Development	Analytics	4	8	running	6.80	6,071.56
vm-0089	stg-moderate-089	gp-4x16	Hyderabad	Staging	Engineering	4	16	running	7.60	8,267.78
vm-0090	dev-temp-idle-090	gp-4x16	Mumbai	Development	QA	4	16	running	7.60	7,652.42
vm-0091	prod-prod-peaks-091	gp-8x32	Mumbai	Production	Data	8	32	running	15.20	17,442.48
vm-0092	prod-prod-peaks-092	gp-8x32	Hyderabad	Production	Data	8	32	running	15.20	17,116.05
vm-0093	dev-dev-office-093	cpu-4x8	Mumbai	Development	QA	4	8	running	6.80	6,952.86
vm-0094	stg-bursty-094	cpu-4x8	Singapore	Staging	Engineering	4	8	running	6.80	9,244.77
vm-0095	prod-batch-095	mem-8x64	Hyderabad	Production	QA	8	64	running	18.40	17,518.64
vm-0096	prod-prod-steady-096	gp-4x16	Mumbai	Production	Engineering	4	16	running	7.60	10,189.87
vm-0097	prod-prod-steady-097	gp-4x16	Hyderabad	Production	Analytics	4	16	running	7.60	11,821.07
vm-0098	dev-steady-low-098	gp-2x8	Hyderabad	Development	Analytics	2	8	running	3.80	4,790.28
vm-0099	stg-moderate-099	cpu-4x8	Singapore	Staging	Platform	4	8	running	6.80	9,135.77
vm-0100	stg-moderate-100	gp-4x16	Hyderabad	Staging	Platform	4	16	running	7.60	10,755.87

4. Utilization data methodology

- Measured fields: CPU percent, memory percent, disk percent, disk IOPS, network in and out (Mbps).
- Sampling interval: 15 minutes. Observation period: 30 days.
- Invalid rows (missing or out-of-range values) are excluded from the analysis.
- Duplicate observations (same resource and timestamp) are counted once; the extra copies are excluded.
- Missing intervals are not filled with zero. They are counted as missing, and completeness is valid measurements divided by expected intervals.
- Where a daily egress value is missing, the cost calculation uses the resource's average daily egress in its place.
- Each resource's observed percentiles, such as p95 and p99 CPU, are calculated from its valid measurements only.

Quality results for the loaded dataset: 286,832 valid measurements, 1,168 missing intervals, 3 invalid rows excluded, 2 duplicate rows excluded.

5. Pricing methodology

- Currency: INR (synthetic). All prices are illustrative synthetic inputs.
- Month convention: 730 hours per month. Hourly compute cost = hourly rate x 730.
- Compute is billed only while a VM runs. Stopping a VM removes its compute cost.
- Storage: INR 6.00 per GB-month. Storage keeps billing while the VM is stopped.
- Public IP: INR 290.00 per month, billed whether or not it is attached (synthetic rule).
- Outbound data transfer (egress): INR 4.50 per GB.
- Off-hours scheduling savings use a schedule active fraction of 0.2976 (scheduled hours divided by 168 hours per week).
- All prices are illustrative synthetic inputs, not cloud-provider quotes.

Assumptions and limitations: these price inputs are not tied to any cloud provider's price list, discount, commitment or region-specific tariff. Savings derived from them are estimates for demonstration, not guaranteed reductions.

6. Data quality and limitations

- Resources with insufficient history: 1. A resource is flagged when completeness is below the minimum threshold or it has too few observed days. Recommendations for flagged resources are given lower confidence or a request for more data.
- Resources below minimum completeness: 1.
- Configuration conflicts: 0.
- Some resources contain demo data-quality faults on purpose (invalid and duplicate rows, and incomplete metrics), so the quality checks can be seen working.
- Optimization recommendations depend on the completeness and history of each resource. Savings are projections, not guaranteed changes to any real infrastructure.

7. Safety disclaimer

The dataset is for demonstration and decision-support purposes. The resources in this report do not represent verified live infrastructure unless an actual cloud integration has been explicitly confirmed. This report does not constitute an actual cloud bill or a guarantee of savings. No recommendation in this application changes any cloud resource.